EVERY FPPF COVERING OF SCHEMES IS A PH COVERING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13787-13790 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13792-13803, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Topology-comparison entry of the chapter's ph section; short, self-contained, fully named.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Topologies, definitions fppf-covering and ph-covering; the chapter's lemma dominate-fppf-finite (finite locally free domination of fppf covers of affines); Morphisms, lemma finite-proper.

Source reference. Original lemma 0CP0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13758–13765. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13803–13803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 021M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13803–13803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DBG: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13803–13803. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *An fppf covering of schemes is a ph covering.*

Proof. Let $\{T_i \to T\}$ be an fppf covering of schemes, see Topologies, Definition 021M. Observe that $T_i \to T$ is locally of finite type. Let $U \subset T$ be an affine open. It suffices to show that $\{T_i \times_T U \to U\}$ can be refined by a standard ph covering, see Topologies, Definition 0DBG. This follows immediately from Lemma 0CP0 and the fact that a finite morphism is proper (Morphisms, Lemma 01WN). □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.